

AWS EC2

What is EC2

EC2 (Elastic Compute Cloud) is AWS's Infrastructure as a Service offering. It lets you rent virtual machines in the cloud.

Core capabilities:

- **EC2** — rent virtual machines
 - **EBS** — store data on virtual drives
 - **ELB** — distribute load across machines
 - **ASG** — scale services using auto-scaling groups
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Instance Configuration

When launching an EC2 instance you configure:

- **OS** — Linux, Windows, or Mac OS
 - **CPU** — compute power and number of cores
 - **RAM** — random-access memory
 - **Storage** — network-attached (EBS, EFS) or hardware (EC2 Instance Store)
 - **Network card** — speed and public IP address
 - **Security group** — firewall rules
 - **EC2 User Data** — bootstrap script that runs once at first launch
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Instance Types

Naming convention: `m5.2xlarge` where `m` = instance class, `5` = generation, `2xlarge` = size

Type	Use Case
General Purpose	Balanced workloads, web servers, code repos
Compute Optimized	Batch processing, ML inference, HPC, gaming servers
Memory Optimized	Large in-memory databases, real-time big data processing
Storage Optimized	OLTP, NoSQL databases, Redis cache, data warehousing

t3.micro is part of the AWS free tier — 750 hours/month

AMI

AMI (Amazon Machine Image) = the OS + pre-installed software template used to launch an instance. You pick one when creating an EC2 instance.

EC2 User Data

A bootstrap script attached to an instance that runs once at first start with root user permissions. Used to automate boot tasks like installing updates, installing software, or downloading files.

Security Groups

Firewall attached to EC2 instances. Controls inbound and outbound traffic via IP and port rules.

- All inbound traffic is blocked by default
 - All outbound traffic is authorised by default
 - Can be attached to multiple instances
 - Scoped to a region and VPC
 - If your app is timing out — it is a security group issue
 - If your app gives connection refused — it is an application error
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Classic Ports

Port	Protocol	Use
22	SSH / SFTP	Log into or transfer files to a Linux instance
21	FTP	Upload files to a file share
80	HTTP	Access unsecured websites
443	HTTPS	Access secured websites
3389	RDP	Log into a Windows instance